

Exercises 09/10

1. $\beta(A, B)$ is the $n \times n$ matrix whose columns are the coordinate representations of the elements of A against B , so that $\beta^v = \beta(A, B)\alpha^v$. Show that $\beta(A, B)^{-1}$ transforms coordinates from B to A .
2. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by

$$T(v_1, v_2, v_3) = \begin{pmatrix} 3v_1 - v_2 \\ v_1 + v_2 + 2v_3 \end{pmatrix}.$$

Find $M(T, \mathcal{E})$, the matrix representation of T against the canonical bases.

3. Let $n = 2$ and T be a linear map represented by

$$M = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}$$

against the canonical basis $\{(1, 0), (0, 1)\}$. Let $B = \{(2, 1), (-2, 1)\}$. Show that B is a basis for $\mathbb{R}^2$ and compute the matrix representation of M against it.

4. Let M be the 3×3 matrix with zeros in every entry except that $M_{12} = M_{13} = 1$. Show that M is not diagonalizable.
5. Show that if M and N are symmetric, positive definite then $M + N$ is too.